SYSDESIGN REFERENCE

# Payment System Capacity Workbook

Traffic, storage, bandwidth, and bottleneck estimation practice.

Move money through an auditable state machine where retries never duplicate financial effects.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / PAYMENT-SYSTEM

SYSDESIGN

# Payment System Capacity Workbook

**CORE IDEA**

**Move money through an auditable state machine where retries never duplicate financial effects.**

## Reference architecture decision chain

| 01 | 02 | 03 | 04 |
|---|---|---|---|
| Ledger | Authorization/Capture | Idempotency | Reconciliation |

### Decisions to defend

+ Ledger
+ authorization/capture
+ idempotency
+ reconciliation
+ webhooks
+ and compliance boundaries.

### Failure modes to test

+ Double charge
+ lost webhook
+ ledger mismatch
+ partial refund
+ and provider outage.

## Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

# Payment System Capacity Workbook

| Probe | What a strong answer makes explicit |
| --- | --- |
| How does the design behave during a regional dependency failure? | Requirement, decision, cost, failure behavior, and observable metric. |
| Which metric reveals the first bottleneck? | Requirement, decision, cost, failure behavior, and observable metric. |
| What consistency does the user actually observe? | Requirement, decision, cost, failure behavior, and observable metric. |
| How would the design change at ten times the traffic? | Requirement, decision, cost, failure behavior, and observable metric. |

## Data questions

- Partition key
- Versioning
- Uniqueness
- Retention
- Reconciliation

## Operations questions

- SLO
- Backlog
- Capacity
- Alert
- Failover test

**DESIGN DISCIPLINE**

**For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?**

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